

Notes for ‘The logistic function’

Important Ideas and Useful Facts:

- (i) **The exponential growth model:** Consider a population of size $x = x(t) > 0$, which is a function of time t . The exponential growth model assumes that the growth rate of x with respect to t is proportional to x , that is,

$$\frac{dx}{dt} = kx$$

for some constant $k > 0$. This makes precise the idea that, the more there is of the population or colony, the faster it should grow. One can show that the only solutions to this *differential equation* (an equation involving a derivative) have the form

$$x = x(t) = Ae^{kt}$$

for some constant $A > 0$, which is the usual rule for exponential growth.

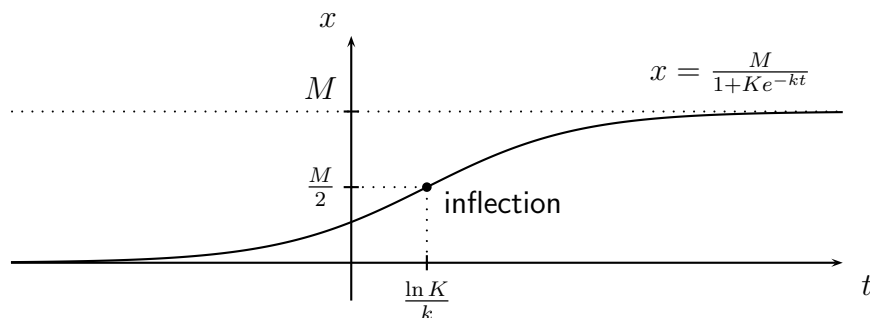
- (ii) **The logistic growth model:** The exponential growth model is modified to take into account constraints on the growth of the population caused by limited resources. A more sophisticated model presumes there is a largest sustainable population size, say $M > 0$, and adjusts the differential equation above for $x = x(t)$, where $k > 0$, by including an *inhibition factor* $(1 - \frac{x}{M})$, which is close to one when x is small, but close to zero when x is close to M :

$$\frac{dx}{dt} = kx\left(1 - \frac{x}{M}\right).$$

This (differential equation) is called the *logistic equation*. Its solutions are called *logistic functions*, and they all have the form

$$x = x(t) = \frac{M}{1 + Ke^{-kt}}$$

for some constant $K > 0$. The graph of x (where x labels the vertical axis, whilst t labels the horizontal axis) has a *sigmoid* shape, concave up followed by concave down. The curve is increasing, with horizontal asymptotes $x = 0$ and $x = M$, and has 180° rotational symmetry about a point of inflection that occurs when $x = \frac{M}{2}$ and $t = \frac{\ln K}{k}$.



Examples and derivations:

1. Find integers A and B such that $\frac{1}{119} = \frac{1}{7 \times 17} = \frac{A}{7} + \frac{B}{17}$.

Solution: We want $1 = 17A + 7B$. We look for multiples of 17 and 7 that differ by 1, say $34 = 2 \times 17$ and $35 = 5 \times 7$, so take $A = -2$ and $B = 5$, and then we have

$$\frac{1}{119} = \frac{-2}{7} + \frac{5}{17}.$$

2. Find constants A and B such that

$$\frac{1}{2x - x^2} = \frac{1}{x(2 - x)} = \frac{A}{x} + \frac{B}{2 - x}.$$

Solution: After rewriting the right-hand side with a common denominator, we find that we need

$$1 = A(2 - x) + Bx = 2A + (B - A)x.$$

To get constants to match, we want $1 = 2A$, so that $A = \frac{1}{2}$. It follows that $B = \frac{1}{2}$ also, so we have

$$\frac{1}{2x - x^2} = \frac{1}{x(2 - x)} = \frac{\frac{1}{2}}{x} + \frac{\frac{1}{2}}{2 - x}.$$

3. Find the indefinite integral $\int \frac{dx}{2 - x}$ (assuming $0 < x < 2$).

Solution: Put $u = 2 - x$, so that $\frac{du}{dx} = -1$. Hence $du = -dx$ and $dx = -du$, and we get

$$\int \frac{dx}{2 - x} = \int \frac{-du}{u} = - \int \frac{du}{u} = -\ln u + C = -\ln(2 - x) + C,$$

noting that $u > 0$, for the third step.

4. Find the indefinite integral $\int \frac{dx}{2x - x^2}$ (assuming $0 < x < 2$).

Solution: Using the solutions to the two previous exercises, we get

$$\begin{aligned} \int \frac{dx}{2x - x^2} &= \int \left(\frac{\frac{1}{2}}{x} + \frac{\frac{1}{2}}{2 - x} \right) dx = \frac{1}{2} \left(\int \frac{dx}{x} + \int \frac{dx}{2 - x} \right) \\ &= \frac{1}{2} \left(\ln x - \ln(2 - x) \right) + C = \frac{1}{2} \ln \left(\frac{x}{2 - x} \right) + C. \end{aligned}$$

5. Consider the logistic function

$$x = x(t) = \frac{M}{1 + Ke^{-kt}},$$

where $M, K, k > 0$. Use interval notation to describe the range and show that the derivative $x'(t)$ is maximised when

$$t = \frac{\ln K}{k},$$

given that this occurs when $x = \frac{M}{2}$.

Solution: The curve is sandwiched between horizontal asymptotes $x = 0$ and $x = M$, so that the range is the interval $(0, M)$. The derivative is maximised when

$$\frac{M}{2} = \frac{M}{1 + Ke^{-kt}} ,$$

so that $2 = 1 + Ke^{-kt}$, giving $Ke^{-kt} = 1$, so that $K = e^{kt}$. Taking natural logarithms and rearranging, we get

$$t = \frac{\ln K}{k} .$$

6. A rumour has been spread, amongst a population of 50,000 people, that calculus is fun. It was seeded by ten people, and after 10 days, it is estimated that 1,000 people know about the rumour. Let $x(t)$ denote the number of people that know about the rumour after t days, and assume a logistic model, so that

$$x = x(t) = \frac{50,000}{1 + Ke^{-kt}}$$

for some positive constants K and k . Find

- (a) exact expressions for K and k ,
- (b) the number of days after which the rumour is being spread most rapidly, and
- (c) the number of days after which 40,000 people know about the rumour.

Solution: (a) We know that

$$10 = x(0) = \frac{50,000}{1 + Ke^0} = \frac{50,000}{1 + K} ,$$

so that $1 + K = 5000$, so that $K = 4999$.

We also know that

$$1,000 = x(10) = \frac{50,000}{1 + Ke^{-10k}} .$$

Hence $1 + Ke^{-10k} = 50$, so that $Ke^{-10k} = 49$, which becomes $e^{10k} = \frac{K}{49}$. Taking natural logarithms and rearranging, we get

$$k = \frac{\ln(K/49)}{10} = \frac{\ln(4999/49)}{10} .$$

- (b) The rumour is being spread most rapidly when

$$t = \frac{\ln K}{k} = \frac{10 \ln(4999)}{\ln(4999/49)} \approx 18.4 ,$$

which is after about 18 days (to the nearest day).

- (c) We want t such that

$$40,000 = \frac{50,000}{1 + Ke^{-kt}}$$

so that $1 + Ke^{-kt} = \frac{5}{4}$, that is, $Ke^{-kt} = \frac{1}{4}$, so that $e^{kt} = 4K$. This becomes

$$t = \frac{\ln(4K)}{k} = \frac{10 \ln(4 \times 4999)}{\ln(4999/49)} \approx 21.4 ,$$

which is after about 21 days (to the nearest day).

7. Suppose a yeast colony starts off with a population of 100 cells and is growing most rapidly 10 hours later, when it reaches 500 cells. Assuming logistic growth, find an upper bound for the colony size, and how long it takes to reach 75% and then 90% of this maximum.

Solution: Let t be the number of hours and $x = x(t)$ be the size of the colony, so that

$$x = \frac{M}{1 + Ke^{-kt}} ,$$

for some constants $M, K, k > 0$.

The population is growing most rapidly after $t = 10$ hours, at which time it reaches 500. This occurs at the point of inflection of the logistic curve, when the population has reached half of its potential maximum, so that $\frac{M}{2} = 500$. Hence, an upper bound for the colony size is $M = 1,000$.

We have

$$100 = x(0) = \frac{M}{1 + Ke^0} = \frac{1000}{1 + K} ,$$

which quickly rearranges to give $K = 9$.

We also have

$$500 = x(10) = \frac{M}{1 + Ke^{-10k}} = \frac{1000}{1 + 9e^{-10k}} ,$$

which quickly rearranges to give

$$e^{-10k} = \frac{1}{9} .$$

Hence $e^{10k} = 9$, so that, taking natural logarithms and rearranging, we get

$$k = \frac{\ln 9}{10} .$$

To reach 75% of the maximum, we need $x = 750$, so that $750 = \frac{1000}{1 + 9e^{-kt}}$, which rearranges to give

$$1 + 9e^{-kt} = \frac{1000}{750} = \frac{4}{3} .$$

Thus $e^{-kt} = \frac{1}{27}$, so that $e^{kt} = 27$. Taking natural logarithms and rearranging, we get

$$t = \frac{\ln(27)}{k} = \frac{10 \ln(27)}{\ln 9} = \frac{10 \ln(3^3)}{\ln(3^2)} = \frac{10 \times 3 \ln 3}{2 \ln 3} = 15 .$$

To reach 90% of the maximum, we need $x = 900$, so that $900 = \frac{1000}{1 + 9e^{-kt}}$, which rearranges to give

$$1 + 9e^{-kt} = \frac{1000}{900} = \frac{10}{9} .$$

Thus $e^{-kt} = \frac{1}{81}$, so that $e^{kt} = 81$. Taking natural logarithms and rearranging, we get

$$t = \frac{\ln(81)}{k} = \frac{10 \ln(81)}{\ln 9} = \frac{10 \ln(9^2)}{\ln 9} = \frac{10 \times 2 \ln 9}{\ln 9} = 20 .$$

Thus we expect to reach 75% of the maximum after 15 hours, and 90% after 20 hours.

8. We explain why the exponential growth model

$$\frac{dx}{dt} = kx ,$$

for some constant $k > 0$, only has solutions of the form $x = Ae^{kt}$ for some constant $A > 0$. We assume throughout that $x = x(t)$ is positive.

Proof: We multiply through by dt to get an equation involving differentials:

$$dx = kxdt$$

and then divide both sides by x to get

$$\frac{dx}{x} = kdt .$$

We then integrate both sides, to get an equation linking antiderivatives:

$$\int \frac{dx}{x} = \int k dt = kt + C ,$$

for some constant of integration C . On the left-hand side, we integrate to get the natural logarithm function, and can suppose that the constant of integration has been moved across and absorbed in the constant on the right, and we get

$$\ln x = kt + C .$$

Raising Euler's number e to the power of each side of this equation, we get

$$x = e^{\ln x} = e^{kt+C} = e^C e^{kt} = Ae^{kt} ,$$

for some constant A , taking $A = e^C$, completing the proof.

9. We explain why the logistic growth model

$$\frac{dx}{dt} = kx \left(1 - \frac{x}{M} \right) ,$$

for some constants $k, M > 0$, only has solutions of the form

$$x = \frac{M}{1 + Ke^{-kt}}$$

for some constant $K > 0$. We assume throughout that $0 < x < M$.

Proof: We multiply through by dt to get an equation involving differentials:

$$dx = kx \left(1 - \frac{x}{M} \right) dt$$

and then divide both sides by $x(1 - \frac{x}{M})$, putting everything involving x to the left-hand side, to get

$$\frac{dx}{x(1 - \frac{x}{M})} = kdt .$$

(This is an example of the technique of *separating variables*, which is explained in greater generality in more advanced courses in calculus.) We then integrate both sides, to get an equation linking antiderivatives:

$$\int \frac{dx}{x(1 - \frac{x}{M})} = \int k dt = kt + C, \quad (1)$$

for some constant of integration C . To unravel the left-hand side, we first observe that

$$\frac{1}{x(1 - \frac{x}{M})} = \frac{1}{x} + \frac{\frac{1}{M}}{1 - \frac{x}{M}} = \frac{1}{x} + \frac{1}{M - x}.$$

It easy to check this equation holds. It is not so clear where it came from. It arises by postulating a decomposition of the left-hand side as a sum of individual fractions with denominators made up from each of the factors of the denominator of the left-hand side, and then using a “guess and check” method. (A more formal, systematic approach uses the *method of partial fractions*, explained in more advanced courses in calculus.)

We then get

$$\int \frac{dx}{x(1 - \frac{x}{M})} = \int \frac{dx}{x} + \int \frac{dx}{M - x} = \ln x - \ln(M - x) + C = \ln\left(\frac{x}{M - x}\right) + C,$$

where the second antiderivative can be deduced quickly by a “guess and check” method or using integration by substitution. Putting this together with (1), by combining the constants of integration on the right, we get

$$\ln\left(\frac{1}{\frac{M}{x} - 1}\right) = \ln\left(\frac{x}{M - x}\right) = kt + C.$$

Raising Euler’s number e to the power of each side of this equation, we get

$$\frac{1}{\frac{M}{x} - 1} = e^{kt+C} = e^C e^{kt},$$

so that, after reciprocating,

$$\frac{M}{x} - 1 = e^{-C} e^{-kt} = K e^{-kt},$$

for some constant K , taking $K = e^{-C}$, which, after rearranging, produces

$$x = \frac{M}{1 + K e^{-kt}},$$

completing the proof.

10. It is worth noting, in the rule above for the logistic function, that when $K e^{-kt}$ is large, then it dominates the fraction, so that the addition of 1 becomes negligible, so that

$$x = \frac{M}{1 + K e^{-kt}} \approx \frac{M}{K e^{-kt}} = \left(\frac{M}{K}\right) e^{kt}.$$

Thus, towards the left of the inflection point of the curve, the logistic function behaves approximately as an exponential function.

This makes sense in terms of the model: for small x , the inhibition factor is close to zero, and the logistic model is approximated well by the exponential growth model. As the population increases, its growth rate is maximised when it reaches $\frac{M}{2}$, after which the population continues to increase, but the growth rate decreases, and dwindles to a trickle the closer the population gets to the maximum carrying capacity M .